

MEMBER 2: DATA ENGINEER - VIBE CODING ROADMAP

Smart India Hackathon 2026 | Team ILLUMINATI (ID: 27113) | PS: SIH1642

Primary Folder: `data/`, `ml/` | Dedicated Branch: `feature/data-engineering` | Deliverables: 3 Files

PHASE 1: INGESTION & METAR PARSING

Prompt 2.1

Create `ml/preprocessing.py`. Write `load_raw_metar(filepath: str) -> pd.DataFrame` that efficiently reads `data/raw/bangalore_2024_metar.csv` handling potential mixed types with `low_memory=False`.

Prompt 2.2

In `ml/preprocessing.py`, extract and clean core columns: map `DATE` to timestamp, temperature to float, dew_point_temperature to float, relative_humidity to float, wind_speed to float, and wind_direction to float.

Prompt 2.3

In `ml/preprocessing.py`, extract atmospheric pressure: parse the altimeter column (`QNH` station pressure in hPa). If missing, fallback to `sea_level_pressure` or `station_level_pressure`.

Prompt 2.4

In `ml/preprocessing.py`, extract precipitation: parse `precipitation_24_hour`, `precipitation_3_hour`, and weather phenomena codes (`pres_wx_MW1` and `REM` for RA, TS, DZ) and derive continuous rainfall in mm.

Prompt 2.5

In `ml/preprocessing.py`, parse timestamps into datetime index. Extract year, month, day, hour, and day_of_year columns.

PHASE 2: SANITY CLEANING & PHYSICAL BOUNDS FILTER

Prompt 2.6

In `ml/preprocessing.py`, implement `validate_physical_bounds(df) -> df`. Reject sensor glitches: Temperature `[-50, 60]` deg C, Humidity `[0, 100]%`, Pressure `[870, 1085]` hPa, Wind Speed `[0, 120]` m/s, Rainfall `>= 0` mm.

Prompt 2.7

In `ml/preprocessing.py`, implement missing value imputation: for gaps `<= 3` hours use forward-fill (`ffill`); for larger gaps, impute using monthly hourly median. Ensure zero NaN values remain.

Prompt 2.8

In `ml/preprocessing.py`, deduplicate records matching the same (location, timestamp). Sort all observations chronologically.

PHASE 3: TEMPORAL & ROLLING FEATURE ENGINEERING

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Prompt 2.9

In `ml/preprocessing.py`, implement cyclical temporal encoding: `sin_day = sin(2*pi*day_of_year/365.25)` and `cos_day = cos(2*pi*day_of_year/365.25)` to represent continuous seasonal cycles.

Prompt 2.10

In `ml/preprocessing.py`, compute diurnal cyclical encoding: `sin_hour = sin(2*pi*hour/24)` and `cos_hour = cos(2*pi*hour/24)`.

Prompt 2.11

In `ml/preprocessing.py`, compute rolling statistics for persistent weather: 7-day rolling mean for temperature, pressure, relative_humidity, and 7-day cumulative sum for rainfall.

Prompt 2.12

In `ml/preprocessing.py`, compute 30-day seasonal moving averages to capture broader climatic shifts.

Prompt 2.13

In `ml/preprocessing.py`, save the cleaned, feature-rich Bengaluru dataset to `data/processed/bangalore_cleaned.csv`.

PHASE 4: MULTI-CITY INDIAN METEOROLOGICAL SYNTHESIZER

Prompt 2.14

Create `ml/data_generator.py`. Define coordinates, elevation, and climatic zones for 10 Indian cities: Bengaluru, Mumbai, Delhi, Chennai, Kolkata, Hyderabad, Ahmedabad, Jaipur, Shimla, and Bhubaneswar.

Prompt 2.15

In `ml/data_generator.py`, generate 10-year historical observations (2015-2025) calibrated to genuine IMD monthly normals (1991-2020) for each city (e.g. Mumbai monsoon peaks in July, Delhi heat in May, Shimla sub-zero winter).

Prompt 2.16

In `ml/data_generator.py`, add realistic diurnal temperature swings (highs at 14:00, lows at 05:00) with inversely correlated relative humidity.

Prompt 2.17

In `ml/data_generator.py`, inject known historical extreme weather benchmarks: 2015 Chennai Cloudburst (Dec 1, 2015), 2022 Northern Heatwave (May 2022 in Delhi), 2020 Cyclone Amphan (May 2020 in Kolkata), 2024 Bengaluru heat record.

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Prompt 2.18

In ml/data_generator.py, tag injected extreme events with an is_ground_truth_anomaly boolean flag for Member 5 to evaluate model accuracy.

Prompt 2.19

In ml/data_generator.py, combine real Bengaluru observations with the 10-city calibrated historical dataset and save to data/processed/weather_cleaned.csv.

Prompt 2.20

Add CLI entrypoints so running python ml/preprocessing.py && python ml/data_generator.py completely rebuilds data/processed/ with clean progress logs.